

Case Statement - Condicional Aggregation SQL TSK

CONDITIONAL AGGREGATION

Apply aggregate functions only on subsets of data that fulfill certain conditions.

SQL TASK

Count how many times each customer has made an order with sales greater than 30.

① Arrume o que você vai usar

```
7 SELECT
8   OrderID,
9   CustomerID,
10  Sales
11 FROM Sales.Orders
12 ORDER BY CustomerID
```

	OrderID	CustomerID	Sales
1	3	1	20
2	4	1	60
3	7	1	30
4	1	2	10
5	5	2	25
6	9	2	20
7	10	3	60
8	6	3	50
9	2	3	15
10	8	4	90

② Flag → Técnica utilizada p/ saber o que atende a condição, no qual você cria uma nova coluna com 0 e 1 (Falso e Verdadeiro)

```
7 SELECT
8   OrderID,
9   CustomerID,
10  Sales,
11  CASE
12    WHEN Sales > 30 THEN 1
13    ELSE 0
14  END SalesFlag
15 FROM Sales.Orders
16 ORDER BY CustomerID
```

→ Botou a flag com o CASE

	OrderID	CustomerID	Sales	SalesFlag
1	3	1	20	0
2	4	1	60	1
3	7	1	30	0
4	1	2	10	0
5	5	2	25	0
6	9	2	20	0
7	10	3	60	1
8	6	3	50	1
9	2	3	15	0
10	8	4	90	1

③ 3.1 → Retirou as colunas OrderID e Sales
3.2 → Agrupou a coluna CustomerID, e somou a coluna SalesFlag e chamou de TotalOrders

```
7 SELECT
8   CustomerID,
9   SUM(CASE
10     WHEN Sales > 30 THEN 1
11     ELSE 0
12   END) TotalOrders
13 FROM Sales.Orders
14 GROUP BY CustomerID
```

	CustomerID	TotalOrders
1	1	1
2	2	0
3	3	2
4	4	1

④ Colocar o Total de Compras de cada cliente

```
7 SELECT
8   CustomerID,
9   SUM(CASE
10     WHEN Sales > 30 THEN 1
11     ELSE 0
12   END) TotalOrders,
13   COUNT(*) Total_Orders_Per_Customers
14 FROM Sales.Orders
15 GROUP BY CustomerID
```

	CustomerID	TotalOrders	Total_Orders_Per_Customers
1	1	1	3
2	2	0	3
3	3	2	3
4	4	1	1

⑤ Note que a Conditional Aggregation é a função de agregação com o CASE